Portfolio Assessment 3

Electrical grid stability classification

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Assisted working analysis. Personal learning reflection and Canvas submission details are deferred by the student.

Question and evaluation protocol

Compare Random Forest, scikit-learn MLP and two required ANN configurations on 11 original grid predictors. The unchanged 10,000-row simulated dataset has 3,620 stable and 6,380 unstable cases. Exclude the label-leaking stab and redundant p1. No missing values, duplicate rows or resampling were found or needed.

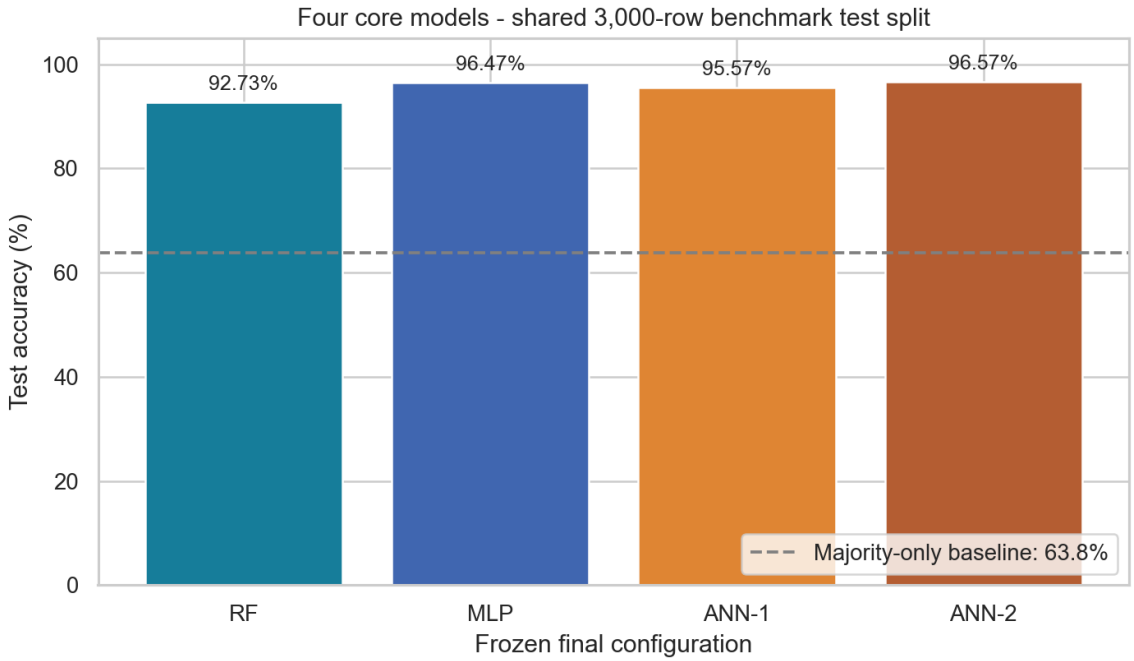
Use the Portfolio 2 stratified 70/30 split (seed 42), a previously inspected benchmark. Traditional models have five-fold stratified development CV with fold-local scaling. ANN trials reserve 1,400 of 7,000 development rows for validation, with a scaler fitted only on the other 5,600. Freeze settings before test inference. All four final models refit on the same common-scaled 7,000 development rows.

Final refit results - shared 3,000-row test split

Model	Accuracy	Weighted F1	Unstable recall	Unstable called stable
RF	92.73%	0.9268	95.92%	78
MLP	96.47%	0.9646	97.39%	50
ANN-1	95.57%	0.9552	98.80%	23
ANN-2	96.57%	0.9655	98.59%	27

ANN-2 has the highest observed accuracy (96.57%). The better core ANN outperformed the better traditional model by +0.10 percentage points. This descriptive test ranking did not choose configurations; small single-split differences are not universal evidence.

MLP has the best balanced accuracy (96.12%); ANN-1 has the fewest unstable-as-stable errors (23). ANN-2 exceeds MLP by only 3 correct test cases.



GitHub: https://github.com/henryhoang207/COS40007-Porfollio-Task

Development evidence

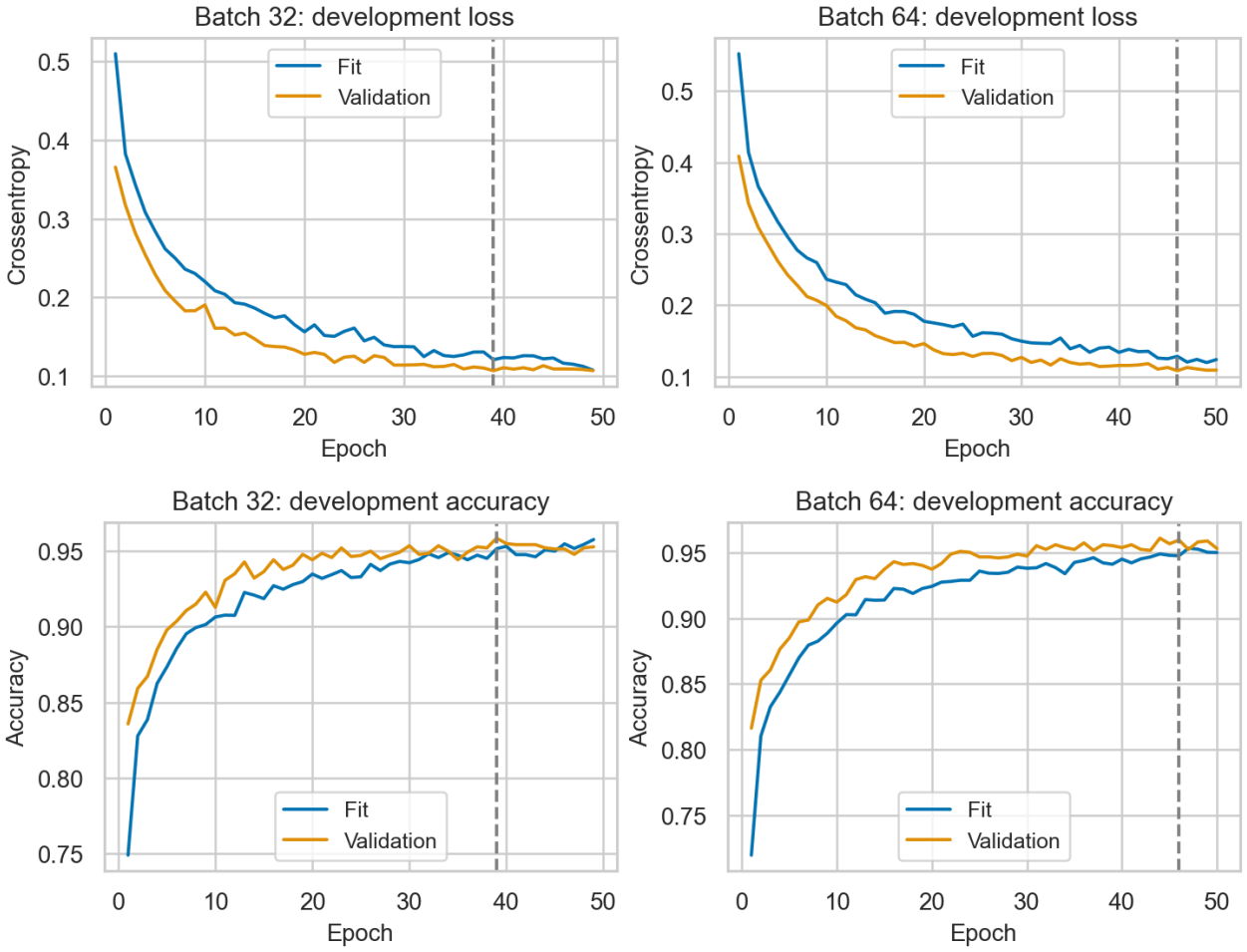
Model	CV accuracy mean	CV accuracy SD	CV weighted F1	CV F1 SD
MLP	0.9541	0.0059	0.9541	0.0059
RF	0.9169	0.0065	0.9163	0.0064

CV SD is sample fold SD (ddof=1), not a confidence interval. ANN five-fold CV was not performed. No traditional MLP convergence warnings occurred at the retained 500-iteration limit.

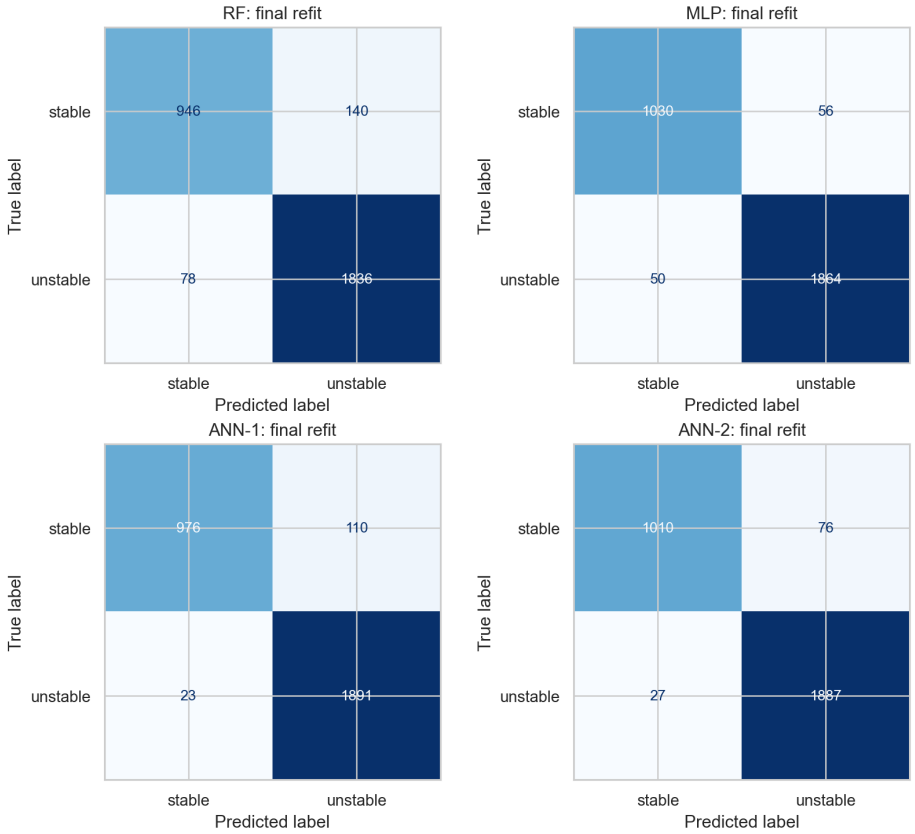
The required ANN has Dense 64/ReLU, dropout 0.2, Dense 32/ReLU, dropout 0.2, Dense 16/ReLU and Softmax(2): 3,410 parameters. Adam starts at 0.001 with sparse categorical crossentropy. A 10-epoch batch-32 pilot precedes three fresh trials capped at 50 epochs. Validation loss controls early stopping (patience 10, best weights restored) and LR halving (patience 5).

Batch	Actual epochs	Best epoch	Best val. loss	Val. accuracy
16	50	40	0.1055	96.07%
32	49	39	0.1070	95.86%
64	50	46	0.1080	95.93%

Core final refits use batch 32 for 39 epochs and batch 64 for 46 epochs, fresh initialisation and constant LR 0.001. Development trial curves below include callbacks and are distinct from the final refits. Batch 16 is retained as a development-only experiment.



Errors and technical reflection



Rows are true class; columns are predicted class. The lower-left entry is an unstable case incorrectly called stable. Per-class precision, recall, F1 and support are retained in the notebook and classification_reports.json.

Architecture choices and comparison

ReLU layers represent nonlinear relationships, dropout regularises training and Softmax matches the two-class target. All widths are fixed, so no best-width or isolated dropout effect is established. Batch size affects update noise and update count per epoch. Both sklearn MLP and Keras ANN are neural networks; implementation, dropout, batch size and stopping differ. These experiments cannot attribute score differences to one factor.

Improvements and deployment

Repeat development trials across seeds; use bounded inner-CV tuning; examine probability calibration, development-selected cost-sensitive thresholds and near-boundary errors. External field validation is needed. Do not deploy this assignment model: simulation-only data, an inspected test split, topology and operating distribution shifts, noise and unequal error costs limit the evidence. Engineering safety checks, monitoring and conservative fallback behaviour remain necessary.

Persistence and reproducibility

Two joblib-compressed .pkl pipelines accept raw ordered features. Two whole-model .h5 files use the saved common scaler; development and final histories are separate .pkl files. A fresh process reloads every model and checks all 3,000 test rows, with exact class agreement and probability atol=rtol=1e-6. It also reconstructs split indices, checks scaler statistics, and recomputes metrics. Use Python 3.12 and requirements-lock.txt; run python run_notebook.py to restart and execute the complete analysis.

GenAI Reflection

1. GenAI Tools Used

OpenAI Codex desktop assistant (GPT-6-based runtime; exact model version not exposed). Used for environment setup, generated implementation, experiments, debugging, technical interpretation, report preparation and GitHub pushes. This disclosure is broader than debugging-only assistance. The brief prohibits whole-code generation; the generated working analysis must not be presented as independently student-authored or as satisfying that restriction.

2. Specific Tasks & Prompts

Task 1: execute the supplied end-to-end plan. Exact user prompt:

"C:\Users\hoanh\OneDrive\Tài liệu\CS material\COS40007\Portfollio_Task3\Portfolio_Task3_End_to_End_Plan.md"

Follow this plan end to end, with verification between each phase and push to github.

Task 2: defer Canvas details and personal learning responses. Exact user reply:

Record this part in .md files as TBD

3. Code Attribution only for debugging and helping find errors

The template heading is retained. Actual assistance includes generated code in every modelling section, execution, independent artifact checks and AI-drafted measured-result interpretations. ASSISTANCE_LOG.md records actual debugging findings. None of this code is claimed to be the student's unaided work.

4. Learning Reflection

The student requested that personal responses remain deferred. The required questions - how GenAI helped learning, what challenges it created and what specific skills developed - are recorded with TBD entries in SUBMISSION_TODO.md. No personal learning experience is invented here. The student must supply and verify those responses before any final submission.

References and submission status

COS40007 Portfolio Assessment 3d brief, pp. 1-6; complete recorded extraction reviewed because the original PDF was absent. Arzamasov, V. (2018), Electrical Grid Stability Simulated Data, UCI, https://doi.org/10.24432/C5PG66 (CC BY 4.0).

Technical sources: https://www.tensorflow.org/install/pip ; https://keras.io/api/models/model_saving_apis/model_saving_and_loading/ ; https://keras.io/2/api/models/model_saving_apis/model_saving_and_loading/ ; https://scikit-learn.org/1.7/modules/generated/sklearn.neural_network.MLPClassifier.html

GitHub: https://github.com/henryhoang207/COS40007-Porfollio-Task

Current Canvas deadline, accepted fields, personal responses and final submission remain in the student-deferred Markdown checklist. No Canvas submission or receipt is claimed.